

National Higher School of Mathematics

Preparatory Cycle
First Year

Academic Year: 2022/2023
Course: Algebra 1

MTE2 Answers

Exercise 1 (11 points).

1. (0,50 pt) $\forall (a,b), (c,d) \in \mathbb{Z}^2$ we have: $(a,b) \otimes (c,d) = (ac, ad+bc) = (ca, cb+da) = (c,d) \otimes (a,b)$. Then $\otimes$ is commutative.
2. (03 pts) Since $(\mathbb{Z}^2, \oplus)$ is a commutative group, then to show that $(\mathbb{Z}^2, \oplus, \otimes)$ is a commutative ring with unity, it is enough to show that $\otimes$ is associative, distributive with respect to $\oplus$, and has a neutral element.

— For any $(a,b), (c,d), (e,f) \in \mathbb{Z}^2$, we have:

$$((a,b) \otimes (c,d)) \otimes (e,f) = (ac, ad+bc) \otimes (e,f) = (ace, acf+ade+bce) \dots (1)$$

and

$$(a,b) \otimes ((c,d) \otimes (e,f)) = (a,b) \otimes (ce, cf+de) = (ace, acf+ade+bce) \dots (2).$$

(1)=(2) then $\otimes$ is associative on $\mathbb{Z}^2$.

— The law $\otimes$ is distributive with respect to $\oplus$, if for any $(a,b), (c,d), (e,f) \in \mathbb{Z}^2$, we have

$$\begin{cases} (a,b) \otimes ((c,d) \oplus (e,f)) = ((a,b) \otimes (c,d)) \oplus ((a,b) \otimes (e,f)) \\ \text{and} \\ ((c,d) \oplus (e,f)) \otimes (a,b) = ((c,d) \otimes (a,b)) \oplus ((e,f) \otimes (a,b)). \end{cases}$$

Since $\otimes$ is commutative it is enough to prove one equality. We have

$$\begin{aligned} (a,b) \otimes ((c,d) \oplus (e,f)) &= (a,b) \otimes (c+e, d+f) \\ &= (a(c+e), a(d+f)+b(c+e)) \\ &= (ac+ae, ad+bc+af+be) \dots (3) \end{aligned}$$

and

$$\begin{aligned} ((a,b) \otimes (c,d)) \oplus ((a,b) \otimes (e,f)) &= (ac, ad+bc) \oplus (ae, af+be) \\ &= (ac+ae, ad+bc+af+be) \dots (4) \end{aligned}$$

(3) = (4) then $\otimes$ is distributive with respect to $\oplus$.

— Let $(e_1, e_2) \in \mathbb{Z}^2$. (e_1, e_2) is a neutral element for $\otimes$ if for all $(a,b) \in \mathbb{Z}^2$ we have $(a,b) \otimes (e_1, e_2) = (e_1, e_2) \otimes (a,b) = (a,b)$. Since the $\otimes$ is commutative we solve only one equation. We have:

$$(a,b) \otimes (e_1, e_2) = (a,b) \iff \begin{cases} ae_1 = a \\ ae_2 + be_1 = b \end{cases} \iff \begin{cases} a(e_1 - 1) = 0 \\ ae_2 + b(e_1 - 1) = 0 \end{cases} \iff \begin{cases} e_1 = 1 \\ e_2 = 0 \end{cases}$$

Then $(1,0)$ is the neutral element for $\otimes$ in $\mathbb{Z}^2$.

Conclusion $(\mathbb{Z}^2, \oplus, \otimes)$ is a commutative ring with unity.

3. (01,50 pt) Zero divisors: Let $(a,b), (c,d) \in \mathbb{Z}^2 \setminus \{(0,0)\}$ such that $(a,b) \otimes (c,d) = (0,0)$. We have:

$$(a,b) \otimes (c,d) = (0,0) \iff \begin{cases} ac = 0 \dots (1) \\ \text{and} \\ ad+bc = 0 \dots (2) \end{cases}$$

From (1): If $a \neq 0$ then $c = 0$, we replace in (2), we get $ad = 0$, thus $d = 0$. Hence $(c,d) = (0,0)$. If $a = 0$ and $b \neq 0$ we get necessarily $c = 0$ then $(0,b) \otimes (0,d) = (0,0)$.

The set of zero divisors is $D = \{(0,b) \mid b \in \mathbb{Z}^*\}$.

4. (02 pts) The invertible elements: $(a, b) \in \mathbb{Z}^2 \setminus \{(0, 0)\}$ is invertible $\iff \exists (c, d) \in \mathbb{Z}^2 \setminus \{(0, 0)\}$ such that $(a, b) \otimes (c, d) = (1, 0)$. we have:

$$(a, b) \otimes (c, d) = (1, 0) \iff \begin{cases} ac = 1 \dots (1) \\ \text{and} \\ ad + bc = 0 \dots (2) \end{cases}$$

(1) gives $a \neq 0$ and $c = \frac{1}{a}$. But $c \in \mathbb{Z}$ implies $a = \pm 1$.

(2) gives $ad = -bc = -\frac{b}{a}$ that means $d = \frac{-b}{a^2}$. Now we distinguish two cases:

— Case 1: if $a = +1$ then $c = 1$ and $d = -b$.

— Case 2: if $a = -1$ then $c = -1$ and $d = -b$.

So the inverse of $(1, b)$ is $(1, -b)$ and the inverse of $(-1, b)$ is $(-1, -b)$. Thus, the set of invertible elements is $U = \{(\pm 1, b) \mid b \in \mathbb{Z}\}$.

5. (01, 50 pt) B is a subring since:

— $(0, 0) \in B$.

— For any $(a, 0), (b, 0) \in B$ we have $(a, 0) \oplus (b, 0) = (a + b, 0) \in B$.

— For any $(a, 0) \in B$, $-(a, 0) = (-a, 0) \in B$.

— For any $(a, 0), (b, 0) \in B$ we have $(a, 0) \otimes (b, 0) = (ab, 0) \in B$.

The subring B is without zero divisors since $B \cap D = \emptyset$.

6. (01 pt) The map f is a ring homomorphism from $(\mathbb{Z}, +, \cdot)$ to $(\mathbb{Z}^2, \oplus, \otimes)$ since:

$\forall n, n' \in \mathbb{Z}$, $f(n + n') = (n + n', 0) = (n, 0) \oplus (n', 0) = f(n) + f(n')$ and $f(nn') = (nn', 0) = (n, 0) \otimes (n', 0) = f(n) \otimes f(n')$.

(01 pt) $\ker(f) = \{n \in \mathbb{Z} \mid f(n) = (0, 0)\} = \{n \in \mathbb{Z} \mid (n, 0) = (0, 0)\} = \{(0, 0)\}$ then f is injective.

(0, 50 pt) The map f is not surjective since $(0, 1) \in \mathbb{Z}^2$ has no preimage.

Exercise 2 (7 points).

1. (02 pts) Showing that $\mathcal{R}$ is an equivalence relation:

— Reflexivity: $\forall (x, y) \in \mathbb{R}^2$, we have $x = 1.x$ and $y = 1.y$ ($a = 1, b = 1$). Then $(x, y) \mathcal{R} (x, y)$.

— Symmetricity: $\forall (x, y), (x', y') \in \mathbb{R}^2$,

$$(x, y) \mathcal{R} (x', y') \iff \exists a, b > 0, x' = ax \text{ and } y' = by \iff x = \frac{1}{a}x' \text{ and } y = \frac{1}{b}y' \iff (x', y') \mathcal{R} (x, y).$$

— Transitivity: $\forall (x, y), (x', y'), (x'', y'') \in \mathbb{R}^2$,

$$(x, y) \mathcal{R} (x', y') \iff \exists a > 0, \exists b > 0, x' = ax \text{ and } y' = by \dots (1)$$

$$(x', y') \mathcal{R} (x'', y'') \iff \exists c > 0, \exists d > 0, x'' = cx' \text{ and } y'' = dy' \dots (2)$$

$$(1) \text{ and } (2) \text{ give: } x'' = cax \text{ and } y'' = bdy \iff (x, y) \mathcal{R} (x'', y'').$$

2. (02 pts)

— $A = (1, 0)$, $\overline{A} = \{(x, y) \in \mathbb{R}^2 \mid (1, 0) \mathcal{R} (x, y)\} = \{(x, y) \in \mathbb{R}^2 \mid \exists a, b > 0, x = a1 \text{ and } y = b0 = 0\}$ then $\overline{A} = \{(a, 0) \mid a > 0\} =]0, +\infty[\times \{0\}$.

— $B = (0, -1)$, $\overline{B} = \{(x, y) \in \mathbb{R}^2 \mid (0, -1) \mathcal{R} (x, y)\} = \{(0, -b) \mid b > 0\} = \{0\} \times]-\infty, 0[$.

— $C = (1, 1)$, $\overline{C} = \{(x, y) \in \mathbb{R}^2 \mid (1, 1) \mathcal{R} (x, y)\} = \{(a, b) \mid a, b > 0\} =]0, +\infty[\times]0, +\infty[$.

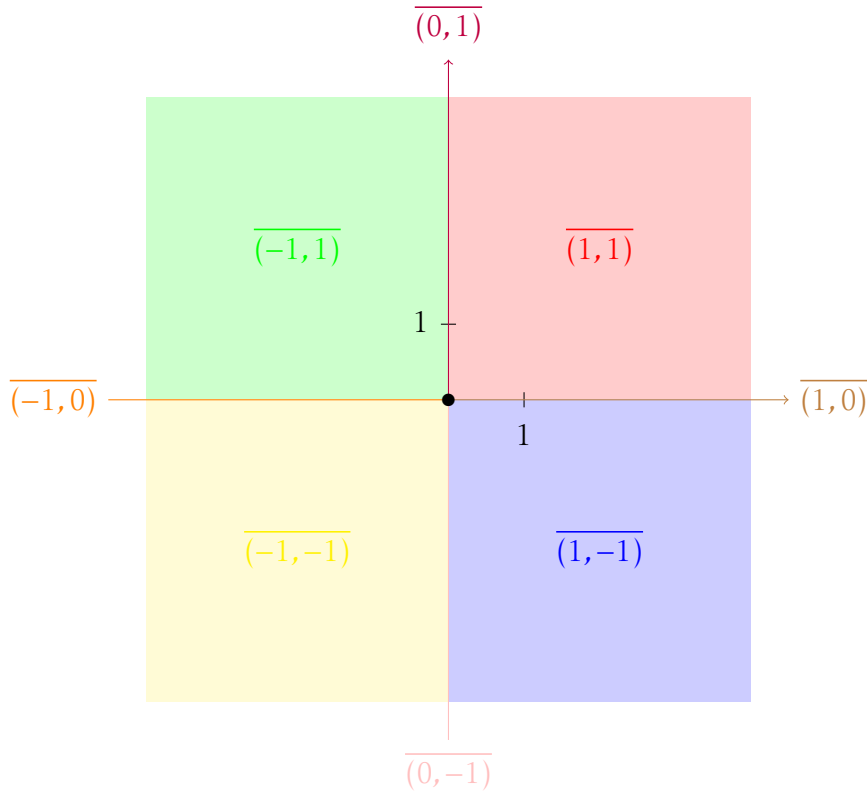
3. (03 pts) Let $(x, y) \in \mathbb{R}^2$. We have $\overline{(x, y)} = \{(x', y') \in \mathbb{R}^2 \mid (x, y) \mathcal{R} (x', y')\} = \{(ax, by) \mid a, b \in]0, +\infty[\}$.

We distinguish 9 cases according to the three possible situations for the value of x : $x > 0$, $x = 0$, $x < 0$ and the three possible situations for the value of y : $y > 0$, $y = 0$, $y < 0$. Thus

$$\mathbb{R}^2 / \mathcal{R} = \{ \overline{(0, 0)}, \overline{(0, 1)}, \overline{(0, -1)}, \overline{(1, 0)}, \overline{(-1, 0)}, \overline{(1, 1)}, \overline{(1, -1)}, \overline{(-1, 1)}, \overline{(-1, -1)} \},$$

where

- $\overline{(0,0)} = \{(0,0)\}$ is the origin,
- $\overline{(1,1)}$ is the right quarter at the top of the plane,
- $\overline{(-1,1)}$ is the left quarter at the top of the plane,
- $\overline{(1,-1)}$ is the right quarter at the bottom of the plane,
- $\overline{(-1,-1)}$ is the left quarter at the bottom of the plane,
- $\overline{(1,0)}$ is the right half of the x-axis, without the origin,
- $\overline{(-1,0)}$ is the left half of the x-axis, without the origin,
- $\overline{(0,1)}$ is the top half of the y-axis, without the origin,
- $\overline{(0,-1)}$ is the bottom half of the y-axis, without the origin.



Exercise 3 (4 points).

1. False. (01 pt)

First Method: If G is a non commutative group, then $\exists a, b \in G$ such that $ab \neq ba$. We have $(ab)^{-1} = b^{-1}a^{-1}$ and $(ba)^{-1} = a^{-1}b^{-1}$. Since G is a group, then the uniqueness of the symmetric element gives us that $(ab)^{-1} \neq (ba)^{-1}$. Hence $(ab)^{-1} \neq a^{-1}b^{-1}$.

Second Method: We can take as a counter-example the non commutative group of the exercise 6 in the worksheet 2. Let

$$\begin{aligned} f: \mathbb{R} \setminus \{0,1\} &\longrightarrow \mathbb{R} \setminus \{0,1\} & g: \mathbb{R} \setminus \{0,1\} &\longrightarrow \mathbb{R} \setminus \{0,1\} \\ x &\longmapsto f(x) = \frac{1}{x} & \text{and} & & x &\longmapsto g(x) = \frac{1}{1-x}. \end{aligned}$$

We have $\forall x \in \mathbb{R} \setminus \{0,1\}$, $(f \circ g)(x) = 1 - x$, $(f \circ g)^{-1} = 1 - x$, $f^{-1}(x) = \frac{1}{x}$, $g^{-1}(x) = \frac{x-1}{x}$ and $(f^{-1} \circ g^{-1})(x) = \frac{x}{x-1}$. Then, it is clear that $(f \circ g)^{-1} \neq f^{-1} \circ g^{-1}$.

2. True. (01 pt) Let $G = \langle a \rangle$, and let $x = a^n$, $y = a^m$. Then $xy = a^{n+m} = a^{m+n} = yx$.

3. **True. (01 pt)** Indeed, let $d = \gcd(n, m)$: if $x \in (n\mathbb{Z} + m\mathbb{Z})$ then $x = ln + mk$, for some $l, k \in \mathbb{Z}$. Hence $x = n'dl + m'dk$, where $n = n'd$ and $m = m'd$. Therefore $x \in d\mathbb{Z}$.

Conversely, If $x \in d\mathbb{Z}$ then $x = dk$ for some $k \in \mathbb{Z}$. In the other hand, using Bezout identity, there exist $a, b \in \mathbb{Z}$ such that $na + mb = d$, thus $n(\underbrace{ak}_{a'}) + m(\underbrace{bk}_{b'}) = dk = x$. Hence, $x \in n\mathbb{Z} + m\mathbb{Z}$.

4. **False. (01 pt)**

First Method: Let $f : (\mathbb{C}^*, \cdot) \rightarrow (\mathbb{R}^*, \cdot)$ be a homomorphism. Then for any $a \in \mathbb{C}^*$, there exists $b \in \mathbb{C}^*$ such that $a = b^2$. Hence, $f(a) = f(b^2) = (f(b))^2 > 0$, thus any negative real number has no preimage. Then f is not surjective. Conclusion: f cannot be an isomorphism.

Second Method: Let $f : (\mathbb{C}^*, \cdot) \rightarrow (\mathbb{R}^*, \cdot)$ be a homomorphism. We have $1 = f(1) = f(i^4) = (f(i))^4$. Since $f(i) \in \mathbb{R}$ then $f(i) = \pm 1$.

— If $f(i) = 1$ we obtain $f(1) = f(i) = 1$, so f is not injective.

— If $f(i) = -1$ we obtain $(f(i))^2 = 1$, so $f(i^2) = f(-1) = 1$. Then $f(1) = f(-1) = 1$, which means that f is not injective.

Conclusion: f cannot be an isomorphism.
